

# Canonical Physical Frame Conditioning and Reassembly Audit:

## Exact Quotient Singular Values, a Sharp Coherence Obstruction, and the Strict Endpoint Gate

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### Abstract

TPC-51 identifies the correct representation space for the inherited Mellin/residue synthesis: the source space must be quotiented by the kernel of the physical synthesis map. TPC-112 then separates the determinant-zero compatibility cone from the strict physical loss ledger. The missing interface is an invariant measurement of the declared quotient conditioning itself and a proof of whether the actual H9 operator chain pays that conditioning.

For one finite literal packet let

$$\mathcal{S}_X : \mathbb{C}^{\Theta_X} \longrightarrow \mathcal{H}_X, \quad \mathcal{S}_X c = \sum_{\theta \in \Theta_X} c_\theta A_{X,\theta},$$

where every column retains its physical coefficient, support and provenance, and let  $\mathcal{K}_X = (\ker \mathcal{S}_X)^\perp$ . For the declared literal source norm, we prove that the exact quotient condition number is

$$\kappa_q(X) = \frac{\sigma_{\max}(\mathcal{S}_X)}{\sigma_{\min}^+(\mathcal{S}_X)} = \sqrt{\frac{\lambda_{\max}^+(G_X)}{\lambda_{\min}^+(G_X)}}, \quad G_X = \mathcal{S}_X^* \mathcal{S}_X.$$

The Moore–Penrose inverse gives the exact optimal recovery constant on  $\text{ran } \mathcal{S}_X$ ; quotienting removes kernel zero-directions, while the positive spectrum still depends on the declared literal presentation. More precisely,  $\kappa_q$  is the sharp uniform envelope

$$\sup_{\|Q\| \leq 1} \|\mathcal{S}_X Q \mathcal{S}_X^\dagger\| = \kappa_q$$

for an arbitrary intervening operator  $Q$  on the quotient source. It is not an inevitable round-trip loss:  $\mathcal{S}_X \mathcal{S}_X^\dagger$  is the identity on the physical range and has norm one. The actual H9 cost is the norm of its literal composite (possibly in a different norm), not  $\kappa_q$  automatically.

We also prove a sharp two-column subframe obstruction without rescaling the physical columns. If their norms are  $\alpha, \beta > 0$  and their normalized coherence is  $\rho$ , the two-column Gram has exact eigenvalues

$$\lambda_{\pm} = \frac{\alpha^2 + \beta^2 \pm \sqrt{(\alpha^2 - \beta^2)^2 + 4\alpha^2\beta^2\rho^2}}{2},$$

and hence

$$\kappa_{q,\text{pair}}^2 = \frac{\lambda_+}{\lambda_-} \geq \frac{1 + \rho}{1 - \rho}.$$

Equality in the lower bound occurs for equal column norms. This obstruction transfers to the complete frame only when the pair is a reducing block, or when a full-Gram Rayleigh certificate proves the same costly direction. Under that explicit inheritance condition,  $1 - \rho \leq X^{-2\chi+o(1)}$  forces the uniform quotient envelope to be at least  $X^{\chi+o(1)}$ . This alone does not stop the strict physical endpoint: a stop additionally requires a lower bound for the actual unavoidable H9 composite or a

proof that it aligns the extremal singular directions. The theorem is exact L0 frame algebra with a conditional L1 bridge. We do not estimate the smallest positive Gram eigenvalue for the growing TPC carrier, obtain an L2 fixed-shift saving, or claim a parity or twin-prime breakthrough.

**Keywords:** canonical synthesis; quotient Gram; physical reassembly; condition number; fixed shift; endpoint obstruction.

**Literal-frame convention.** The column set below is the declared physical column set after exact provenance aggregation. A column may not be rescaled, duplicated or deleted merely to improve a frame constant. One prescribed  $h_0 \neq 0$ , the actual active support and the inherited global normalization remain fixed throughout.

## 1 The inherited interface

TPC-51 constructs a canonical synthesis from residue/Mellin provenance and proves that a lower frame bound on the full source space is impossible whenever the source has a nontrivial kernel [4]. Its correct reduced space is

$$\mathcal{K}_X = (\ker \mathcal{S}_X)^\perp = \text{ran } \mathcal{S}_X^*.$$

The actual coefficient is a distinguished physical direction, not a freely optimized representation. Related TPC-49–TPC-52 results show why an arbitrary projective presentation or a fixed-complexity subpacket cannot replace the inherited full synthesis [3, 5].

TPC-112 proves that any physical cost is independent of the determinant reserve unless an explicit map-specific intertwining is supplied [2]. The present paper measures one candidate round-trip condition number. It does not identify that number with the forward H9 synthesis norm, import a determinant exponent, or spend determinant reserve.

## 2 Canonical quotient cost

Let  $\Theta$  be a finite provenance set, let  $\mathcal{H}$  be a finite-dimensional Hilbert space, and let

$$\mathcal{S} : \mathbb{C}^\Theta \rightarrow \mathcal{H}, \quad \mathcal{S}c = \sum_{\theta \in \Theta} c_\theta A_\theta.$$

Zero columns are allowed, because an inherited mask may delete a source coordinate. Put

$$G = \mathcal{S}^* \mathcal{S}, \quad \mathcal{K} = (\ker \mathcal{S})^\perp.$$

**Definition 2.1** (Declared quotient and recovery constants). If  $\mathcal{S} = 0$ , set both costs to zero. Otherwise define

$$C_{\text{rec}}(\mathcal{S}) = \|\mathcal{S}^\dagger\| = \frac{1}{\sigma_{\min}^+(\mathcal{S})}$$

and

$$\kappa_{\text{q}}(\mathcal{S}) = \|\mathcal{S}\| \|\mathcal{S}^\dagger\| = \frac{\sigma_{\max}(\mathcal{S})}{\sigma_{\min}^+(\mathcal{S})}.$$

Here  $^\dagger$  is the Moore–Penrose inverse and  $+$  means the smallest positive singular or spectral value.

**Theorem 2.2** (Exact quotient-frame reassembly). *For every  $c \in \mathcal{K}$ ,*

$$\sigma_{\min}^+(\mathcal{S}) \|c\|_2 \leq \|\mathcal{S}c\|_{\mathcal{H}} \leq \sigma_{\max}(\mathcal{S}) \|c\|_2.$$

*For every  $y \in \text{ran } \mathcal{S}$ , the unique least-norm coefficient is  $c_y = \mathcal{S}^\dagger y$ , and*

$$\|c_y\|_2 \leq C_{\text{rec}}(\mathcal{S}) \|y\|_{\mathcal{H}}.$$

*Both constants are optimal. If  $\mathcal{S} \neq 0$ , moreover*

$$\boxed{\kappa_{\text{q}}(\mathcal{S})^2 = \frac{\lambda_{\max}^+(G)}{\lambda_{\min}^+(G)}}.$$

*Proof.* Restrict  $\mathcal{S}$  to  $\mathcal{K}$ . It is injective and has the same nonzero singular values as  $\mathcal{S}$ . The singular-value decomposition gives the two-sided inequality and the least-norm inverse. Singular vectors at the extremal singular values show optimality. The positive eigenvalues of  $G = \mathcal{S}^* \mathcal{S}$  are the squares of the nonzero singular values.  $\square$

*Remark 2.3* (Why the quotient is compulsory). The raw Gram always has a zero eigenvalue when exact redundant coordinates are retained. Declaring its ordinary condition number infinite confuses a representation kernel with a physical obstruction. Conversely, silently deleting a nearly dependent but nonredundant column can erase real conditioning. The positive spectrum is canonical only relative to the declared literal source metric. It is invariant under unitary source changes, but not under arbitrary redundant expansions or nonunitary reparameterizations. For example,  $I_2$  has quotient condition number 1, whereas the same physical range presented by columns  $[e_1, e_1, e_2]$  has positive singular values  $\sqrt{2}, 1$  and quotient condition number  $\sqrt{2}$ .

### 3 Uniform conjugation envelope and the actual H9 norm

**Proposition 3.1** (Sharp uniform conjugation envelope). *Assume  $\mathcal{S} \neq 0$ . For operators  $Q : \mathcal{K} \rightarrow \mathcal{K}$ ,*

$$\sup_{\|Q\| \leq 1} \|\mathcal{S} Q \mathcal{S}^\dagger\|_{\text{ran } \mathcal{S} \rightarrow \text{ran } \mathcal{S}} = \kappa_q(\mathcal{S}).$$

*Consequently  $\|\mathcal{S} Q \mathcal{S}^\dagger\| \leq \kappa_q \|Q\|$ , and the corresponding uniform squared-norm envelope is  $\kappa_q^2 \|Q\|^2$ . However,*

$$\mathcal{S} I_{\mathcal{K}} \mathcal{S}^\dagger = I_{\text{ran } \mathcal{S}},$$

*so the identity intermediate operator has amplitude cost one, not  $\kappa_q$ . If*

$$\kappa_q(X) = X^{\chi+o(1)},$$

*then  $\chi$  (or  $2\chi$  for squared norms) is a safe uniform envelope only for a matching composite; it is not an automatic H9 token.*

*Proof.* Submultiplicativity gives the upper bound. Let  $v_{\min}, v_{\max} \in \mathcal{K}$  be right singular vectors for  $\sigma_{\min}^+$  and  $\sigma_{\max}$ , and take the norm-one rank-one operator  $Q v_{\min} = v_{\max}$ , zero on  $v_{\min}^\perp$ . Applied to the corresponding left singular vector,  $\mathcal{S} Q \mathcal{S}^\dagger$  has norm  $\sigma_{\max}/\sigma_{\min}^+ = \kappa_q$ , proving sharpness. The identity formula is the Moore–Penrose range-projection identity.  $\square$

*Remark 3.2* (Actual direction versus uniform frame). TPC-51 also defines an actual-direction ratio. A good bound for that one direction can be sufficient if every subsequent operation preserves it exactly. A theorem applied uniformly to a family must instead use the reduced Gram on the declared family. This paper does not replace one quantifier by the other.

*Remark 3.3* (Not automatically the H9 norm). The MVP1 endpoint controls a literal map from packet amplitudes to a scalar, naturally with a max-packet input norm. Depending on the proved factorization, that map may pay only  $\|\mathcal{S}\|$ , only  $\|\mathcal{S}^\dagger\|$ , the actual composite  $\|\mathcal{S} Q \mathcal{S}^\dagger\|$ , or a different  $\ell^\infty$ -to-scalar norm. The uniform bound  $\kappa_q \|Q\|$  can certify an upper cost, but a large  $\kappa_q$  does not certify a lower cost. No entry may be placed in the H9 ledger until that operator chain and its norm conversions are displayed.

### 4 A sharp physical coherence obstruction

For each nonzero physical column write

$$\alpha_\theta = \|A_\theta\|, \quad u_\theta = \frac{A_\theta}{\alpha_\theta}.$$

The directions are used only to define coherence; the synthesis itself retains the literal column norms.

**Theorem 4.1** (Exact two-column obstruction). *Let two declared nonzero physical columns have norms  $\alpha, \beta > 0$  and normalized coherence*

$$\rho = |\langle u_\theta, u_{\theta'} \rangle| < 1.$$

*Let  $\mathcal{S}_{\text{pair}} : \mathbb{C}^2 \rightarrow \mathcal{H}$  be the synthesis consisting of precisely these two literal columns. Put*

$$\Delta = \sqrt{(\alpha^2 - \beta^2)^2 + 4\alpha^2\beta^2\rho^2}.$$

*Then*

$$\lambda_{\pm}(\mathcal{S}_{\text{pair}}^* \mathcal{S}_{\text{pair}}) = \frac{\alpha^2 + \beta^2 \pm \Delta}{2}$$

*and*

$$\boxed{\kappa_{\text{q}}(\mathcal{S}_{\text{pair}})^2 = \frac{\alpha^2 + \beta^2 + \Delta}{\alpha^2 + \beta^2 - \Delta} \geq \frac{1 + \rho}{1 - \rho}.}$$

*The inequality is an equality if and only if  $\alpha = \beta$ .*

*Proof.* After multiplying one column by a unit phase, the principal two-column literal Gram is

$$\begin{pmatrix} \alpha^2 & \alpha\beta\rho \\ \alpha\beta\rho & \beta^2 \end{pmatrix},$$

whose eigenvalues are the displayed  $\lambda_{\pm}$ . Moreover

$$\Delta^2 - \rho^2(\alpha^2 + \beta^2)^2 = (1 - \rho^2)(\alpha^2 - \beta^2)^2 \geq 0.$$

Thus  $\Delta/(\alpha^2 + \beta^2) \geq \rho$ , which gives the ratio bound after applying the increasing function  $t \mapsto (1+t)/(1-t)$ . Equality holds exactly when  $\alpha = \beta$ .  $\square$

**Corollary 4.2** (Polynomial near-collinearity envelope). *Suppose an actual pair in the declared physical family satisfies*

$$1 - \rho_X \leq X^{-2\chi+o(1)},$$

*and suppose either:*

- (i) its two-dimensional source coordinate space is a reducing block of the full Gram; or*
- (ii) an independent full-Gram certificate proves  $\kappa_{\text{q}}(\mathcal{S}_X) \geq \kappa_{\text{q}}(\mathcal{S}_{\text{pair},X})$ .*

*then*

$$\kappa_{\text{q}}(X) \geq X^{\chi+o(1)}.$$

*This quotient lower bound alone is not a route stop. If a separate literal certificate proves that the actual unavoidable H9 composite has amplitude norm at least  $X^{\chi+o(1)}$ , for example by forced alignment of the relevant singular directions, then  $\chi \geq 1/400$  is incompatible with the strict physical endpoint for that route.*

*Proof.* By [theorem 4.1](#), the pair quotient condition number is at least  $(1 - \rho_X)^{-1/2}$ . Either inheritance hypothesis transfers this condition number to the complete frame. The extra actual-composite lower bound then invokes the strict endpoint rule of TPC-MVP1 [\[1\]](#).  $\square$

**Remark 4.3** (A lower bound must be literal). An abstract pair of near-collinear vectors is only an L0 countermodel. Adding further columns does not generically preserve the condition number of a principal two-column Gram block: the pair direction can mix with the kernel or the rest of the positive spectrum. The exact two-column formula above retains the inherited column norms, but a physical full-frame stop certificate must additionally prove one of the inheritance hypotheses. No such growing full-frame certificate is produced here.

## 5 Finite exact regression certificate

The script `experiments/tpc113_frame_certificate.py` uses exact rational arithmetic to check:

1. Gram and Moore–Penrose identities for several rectangular full-row-rank synthesis matrices;
2. the quotient projection and its idempotence;
3. the nonzero singular-value ledger through  $SS^*$ ;
4. the exact equal-norm two-column eigenvalues  $1 \pm \rho$  for rational Pythagorean pairs;
5. the sharp condition-number ratio  $(1 + \rho)/(1 - \rho)$ ; and
6. the identity-intermediate cost one and a swap intermediate that attains the sharp uniform conjugation envelope.

The committed JSON is a finite algebra certificate. It is not numerical evidence for a lower frame bound on the growing TPC coefficient.

## 6 Stop/go verdict and claim boundary

| Status          | Required certificate                                                                                                                                                             | Consequence                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Go              | A proved actual H9 composite/norm upper bound below the remaining budget; $\kappa_q \ Q\ $ may be used as a uniform upper certificate for a matching $SQS^\dagger$ factorization | Charge the exponent of the proved actual upper bound exactly once, then continue to residual grouping. |
| Stop this frame | An actual unavoidable pair or subspace forces a quotient obstruction and a separate certificate lower-bounds the actual H9 composite by $X^{\chi+o(1)}$ with $\chi \geq 1/400$   | The strict physical endpoint fails for this frame; seek a different lossless representation.           |
| Open            | Only finite spectra, model frames or selected directions are known                                                                                                               | Do not assign exponent zero.                                                                           |

**Established.** Theorems 2.2 and 4.1 and proposition 3.1 give an exact declared quotient condition number, its sharp uniform conjugation envelope and a sharp coherence obstruction. The finite algebra is L0; attachment to H9 is only a conditional L1 bridge requiring the actual operator factorization and norm conversion.

**Not established.** No asymptotic lower bound for the smallest positive physical Gram eigenvalue is proved. No actual polynomial obstruction is exhibited. No equality between  $\kappa_q$  and the H9 forward synthesis or actual composite norm is asserted, and no lower bound for the actual H9 composite follows from a large quotient condition number. No residual grouping, native-Gram comparison, fixed- $h_0$  localization, high-frequency return or full-block reassembly is closed. There is no L2 fixed-power arithmetic saving, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound or twin-prime theorem. Nothing is specialized to  $h_0 = 2$ .

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